

Demonstration of Proposed Features

Date	10 July 2026
Team ID	XXXXXX
Project Name	Raising Waters
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Rainfall Depth Monitoring	Interface to input and process cumulative rainfall in mm over specified periods for ML analysis.	Implemented	Yes	Fully operational; used as a primary feature vector.
2	Cloud Coverage Detection	Ability to ingest and process percentage cloud cover values to assess meteorological complexity.	Implemented	Yes	Key meteorological predictor feature integrated into the model.
3	Historical Data Logging	System to record past rainfall, cloud, and corresponding flood levels into a structured local database for re-training.	Partial	Yes	Recording works, but database re-indexing requires periodic maintenance.
4	Offline Local Inference	Execute the flood risk classification model locally using array data, eliminating external API dependencies.	Implemented	Yes	Core architecture requirement, model weights (.pkl) are integrated and loaded at runtime.
5	Web Interface Dashboard	Responsive web dashboard to input data, view risk results, and monitor system status.	Implemented	Yes	Clean, color-coded status alerts; interface is fully dynamic.
6	Automated Sensor Ingestion	Interface for real-time, automated data stream ingestion from localized weather stations.	Pending	No	Integration block is designed, but hardware-specific APIs are yet to be mapped.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	4 (Full) + 1 (Partial) = 5
Total Features Demonstrated	5
Overall Implementation Rate (%)	$(5/6) \times 100 = 83.33\%$